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18CSP107L-MINOR PROJECT

WINDMILL PITCH PREDICTION AND ERROR CORRECTION WITH ADAPTIVE CONTROL AND CURRENT GENERATION ANALYSIS USING DATA ANALYSIS

BATCH NUMBER: 5

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1

AGENDA

- Abstract
- Scope and Motivation
- Introduction
- Literature Survey (Table)
- Objective
- Problem Statement
- Proposed Work
  - Architecture Diagram/Flow Diagram/Block Diagram
  - Novel idea
  - Modules
  - Module Description
- Software & Hardware Requirements
- References (Base paper hard copy to be submitted to the supervisor.)
- Way forward toward Outcome (Research Paper/Patent)

ABSTRACT

This research paper proposes a comprehensive approach for enhancing windmill pitch control by integrating adaptive control and current generation analysis, emphasizing data analysis techniques. To improve wind turbine performance and energy capture efficiency, this research paper will leverage data-driven methodologies to predict the optimal pitch angle and implement real-time error correction. The research harnesses adaptive control strategies to achieve accurate pitch angle predictions, including Model Predictive Control (MPC) and Gain Scheduling Control. These techniques leverage dynamic wind turbine models and adapt to changing wind conditions, optimizing pitch angles to maximize energy extraction from the available wind resources.

SCOPE AND MOTIVATION

- The scope of this research encompasses simulations and experimental validations to evaluate the proposed approach's effectiveness in terms of wind turbine efficiency, energy production, and cost-effectiveness. The findings aim to contribute to advancing wind energy technology, making it more reliable, scalable, and economically viable.
- The motivation behind this research lies in pursuing sustainable energy solutions, with wind energy playing a pivotal role. By integrating adaptive control strategies and data analysis with current generation analysis, this study seeks to optimize windmill pitch control and unlock the full potential of wind resources, thus paving the way for a greener and cleaner energy future.

INTRODUCTION

The efficient harnessing of wind energy hinges upon optimal wind turbine performance, particularly in predicting and adjusting the windmill pitch. Our paper, "Windmill Pitch Prediction and Error Correction with Adaptive Control and Current Generation Analysis Using Data Analysis," addresses this challenge. This study uses modern data analysis techniques to explore the correlation between windmill pitch adjustments and power output. We aim to enhance wind turbine efficiency and power generation by marrying adaptive control mechanisms with pitch predictions. The subsequent sections elucidate our methodology, insights, and potential impact on the future of wind energy.

LITERATURE SURVEY

S.NO	TITLE OF THE PAPER	YEAR	JOURNAL/CONFERENCE NAME	INFERENCE
1	Fuzzy Logic-Based Pitch Angle Control for Variable Speed Wind Systems	2023	9th International Conference on Advanced Computing and Communication Systems	The paper explores the application of fuzzy logic for optimizing pitch angle control in variable-speed wind systems, leading to enhanced turbine efficiency and adaptability to changing wind conditions.
2	Integrated wind energy conversion and storage for improved electricity quality (WECSS)	2008	IEEE Power Energy Soc. Gen. Meeting	The paper likely investigates the integration of wind energy conversion systems with energy storage solutions, aiming to enhance the quality and reliability of electricity generated from wind sources.

LITERATURE SURVEY

S.NO	TITLE OF THE PAPER	YEAR	JOURNAL/CONFERENCE NAME	INFERENCE
3	Output power levelling of wind turbine generator for all operating regions by pitch angle control	2006	IEEE Trans. Energy Conference	This paper explores the use of pitch angle control to achieve consistent output power from wind turbines across all operational regions, enhancing overall performance and reliability.
4	Analysis of dynamic behavior of electric power systems with large amount of wind power	2003	Ph.d dissertation	The dissertation delves into the dynamic behaviors of electric power systems with significant wind power contributions, assessing their stability and performance impacts.

LITERATURE SURVEY

S.NO	TITLE OF THE PAPER	YEAR	JOURNAL/CONFERENCE NAME	INFERENCE
5	Control of Overload Safety in Wind Turbines Through Blade Pitch Control Implementing Artificial Intelligence	2021	Conference on Electrical, Computer and Energy Technologies	This paper discusses using artificial intelligence in blade pitch control to manage and prevent overloads in wind turbines, enhancing safety and operational efficiency.
6	Fuzzy pitch angle control of wind hybrid turbine: To power quality improvement	2013	XXIV International Conference on Information, Communication and Automation Technologies	This paper investigates the application of fuzzy logic in pitch angle control for wind hybrid turbines, aiming to enhance power quality and system stability.

LITERATURE SURVEY

S.NO	TITLE OF THE PAPER	YEAR	JOURNAL/CONFERENCE NAME	INFERENCE
7	Output power leveling of wind turbine generator by pitch angle control using adaptive control method	2004	International Conference on Power System Technology, 2004. PowerCon	This paper explores adaptive control methods for pitch angle control in wind turbines, aiming to stabilize and level the output power across various conditions.
8	Output power leveling of wind turbine generators using pitch angle control for all operating regions in wind farm	2005	13th International Conference on, Intelligent Systems Application to Power Systems	This paper discusses using pitch angle control in wind turbines within wind farms to ensure consistent output power across all operational regions, optimizing energy yield.

LITERATURE SURVEY

S.NO	TITLE OF THE PAPER	YEAR	JOURNAL/CONFERENCE NAME	INFERENCE
9	Method for Image Recognition and Processing of Windmill Contour	2023	IEEE 3rd International Conference on Power, Electronics and Computer Applications	Proposes a novel technique for identifying and processing the silhouette or outline of windmills in images, potentially aiding in automation or monitoring tasks related to wind energy infrastructure.
10	An assessment of potential windmills using image processing and artificial intelligence AI base	2005	International Conference on Energy, Communication, and Data Analytics	The paper investigates the use of image processing and AI to evaluate the potential or efficiency of windmills, possibly determining optimal locations, assessing structural integrity, or predicting performance based on visual data.

LITERATURE SURVEY

S.NO	TITLE OF THE PAPER	YEAR	JOURNAL/ CONFERENCE NAME	INFERENCE
11	Design &Implementation of an Efficient Windmill Anemometer for Wind Speed measurement Using Microcontroller	2016	International Conference on Energy consumption	The paper explores how image processing combined with AI can enhance the assessment of windmill efficiency and potential. This includes optimizing placement, checking physical condition, and forecasting operational outcomes from visual inputs.
12	Design of Off-Center Fed Windmill Loop for Pattern Reconfiguration	2019	IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS	The paper delves into a novel design of a windmill loop antenna that is fed off-center, aiming to achieve reconfigurable radiation patterns. This design likely offers flexibility and improved performance in specific wireless communication or radar scenarios.

LITERATURE SURVEY

S.NO	TITLE OF THE PAPER	YEAR	JOURNAL/CONFERENCE NAME	INFERENCE
13	Harvesting Renewable Energy through Maglev Windmill	2022	International Interdisciplinary Conference on Mathematics, Engineering and Science	The paper explores the integration of magnetic levitation (Maglev) technology with windmills to harness renewable energy. This approach likely offers reduced friction, increased efficiency, and longer operational lifespans for wind turbines.
14	High-Gain Windmill-Shaped Circularly Polarized Antenna Using the High-Order Mode and Ground-Edge Diffraction	2018	IEEE ANTENNAS AND WIRELESS PROPAGATION LETTERS	The paper presents a windmill-shaped antenna that utilizes high-order modes and ground-edge diffraction to achieve high-gain and circularly polarized signals, potentially enhancing signal strength and clarity in specific communication or radar applications.

LITERATURE SURVEY

S.NO	TITLE OF THE PAPER	YEAR	JOURNAL/CONFERENCE NAME	INFERENCE
15	Wireless Sensor Network for Monitoring Windmills at State Polytechnic of Malang	2021	International Conference on Electrical and Information Technology	The paper describes the implementation of a wireless sensor network at State Polytechnic of Malang to monitor the performance and conditions of windmills, providing real-time data collection and potentially enhancing maintenance efficiency and operational reliability.

OBJECTIVE

This research aims to develop an innovative and data-driven approach for optimizing windmill pitch control, leveraging the integration of adaptive control techniques and current generation analysis. This study aims to achieve accurate pitch angle predictions and implement real-time error correction to enhance wind turbine performance and energy capture efficiency by harnessing data analysis methodologies.

PROBLEM STATEMENT

The effectiveness of wind energy harvesting is substantially dependent on the precision of windmill pitch settings. An inaccurate pitch can lead to diminished power output, increased wear and tear on turbine components, and suboptimal utilization of available wind resources. Despite the technological advancements in wind turbine design, there remains a persistent challenge in predicting the optimal windmill pitch in real-time, accounting for ever-changing wind conditions. Moreover, when pitch predictions err, swift and effective correction mechanisms are only sometimes in place, leading to prolonged periods of inefficiency. Current control systems often operate on predefined settings, needing more adaptability to instantaneous data. The necessity for an integrated approach, which combines real-time windmill pitch prediction, adaptive error correction, and continuous current generation analysis using sophisticated data analysis techniques, is evident. Addressing this issue can pave the way for maximizing wind turbines' efficiency, longevity, and economic viability.

PROPOSED WORK-FLOW DIAGRAM

1.WIND COLLECTION DATA:
-Sensors
-Time-stamped data

2.DATA PREPROCESSING:
-Data Cleaning
-Data Transformation
-Feature Extraction

3.PITCH PREDICTION:
-Inputting processed wind data
-Predictive model of machine learning
-Predicted Optimal pitch angle

4.CURRENT GENERATION ANALYSIS:
-Analyze the relationship between different pitch angles.

5.ERROR DETECTION & CORRECTION:
-After the difference in wind data taken
-Error correction algorithm using adaptive controllers
-This gives the controller corrected pitch angle commands with current generation analysis

6.ADAPTIVE CONTROL:
-Compare generation with predicted pitch angle with optimal pitch
-Adjust pitch angle according to generation profit

7.DATA ANALYSIS & FEEDBACK
-Analyze, Refine modes using a feedback loop

PROCESSED WORK – NOVEL IDEA

This concept integrates data analysis to optimize wind turbine performance. The system uses historical and real-time wind data to predict the best blade angle (pitch) for imminent conditions. If real-world conditions differ from predictions, adaptive controls adjust in real-time. Simultaneously, the system monitors the electricity generated, ensuring the adjustments harness wind efficiently and produce stable, quality power. This is a smart, self-optimizing wind turbine system powered by data analytics.

PROCESSED WORK – MODULE AND MODULE DESCRIPTION

Modules and Descriptions:

- **Wind Data Collection Module:** Collects real-time and historical wind data via sensors.
- **Pitch Prediction Module:** Uses data to forecast the optimal blade pitch for upcoming conditions.
- **Adaptive Control and Error Correction Module:** Adjusts blade pitch in real-time based on discrepancies between predicted and actual wind conditions.
- **Current Generation Monitoring Module:** Tracks the quality and quantity of turbine-generated electricity.
- **Data Analysis and Feedback Loop Module:** Analyzes data to refine predictive algorithms and enhance system performance.

PROCESSED WORK – MODULE AND MODULE DESCRIPTION

- **User Interface and Reporting Module:** Allows monitoring and manual intervention and provides performance reports.
- **Safety and Anomaly Detection Module:** Monitors for unexpected events and triggers safety protocols when necessary.

These modules collaborate to optimize wind turbine performance using data-driven strategies and adaptive controls.

SOFTWARE AND HARDWARE REQUIRMENTS

Hardware Level Requirements:

- **Wind Turbine Sensors:**
  - Anemometers: Measure wind speed and direction.
  - Gyroscope and Accelerometer: Measure and monitor the turbine's movement and vibration.
- **Pitch Angle Sensors:** Detect the current angle of the blades.
- **Data Acquisition Systems (DAQ):** To collect data from various sensors and convert analog signals to digital for processing.
- **Adaptive Controllers:** Capable of adjusting the pitch in real-time based on the received signals and the processing output.
- **Current Generation Meters:** To continuously measure the turbine's power output.

SOFTWARE AND HARDWARE REQUIRMENTS

- **Communication Modules:** Enables remote monitoring and control, especially for wind farms spread across large areas. This could include IoT modules for real-time data transmission.
- **Dedicated Servers/Computational Units:** For data storage, analysis, and running the prediction algorithms, especially if processing on-site.

Software Level Requirements:

- **Real-time Operating System (RTOS):** Essential for applications that require immediate response like real-time pitch adjustments. We are using Scada

SOFTWARE AND HARDWARE REQUIRMENTS

- **Data Analysis Software:**
  - Time-series Analysis Tools: Given the sequential nature of the data. We are using Pandas
  - Machine Learning Frameworks: TensorFlow for developing and deploying prediction models.
- **Database Management Systems (DBMS):** For storing the vast amounts of data collected, We are using PostgreSQL
- **Simulation Software:** MATLAB for initial testing and model validation.
- **Control Algorithm Development Environment:** To design, test, and deploy adaptive control algorithms. We are using MATLAB with Simulink.

SOFTWARE AND HARDWARE REQUIRMENTS

- **Data Visualization Tools:** Tableau for visual data representation, prediction outputs, and system performance metrics.
- **Communication Protocols:** MQTT for IoT-based communication between sensors, controllers, and servers.
- **Cybersecurity Protocols:** Given the critical nature of wind turbine operations, software that ensures data integrity and prevents unauthorized access is crucial.
- **Cloud Computing Services:** For remote processing and storage, especially if data is to be analyzed off-site or in large wind farm setups. We are using GCP.

REFRENCES

BASE PAPER:

G. P. R. Reddy, T. U. Chnadrakanth, V. Sowmyasree and D. Vannurappa, "Fuzzy Logic-Based Pitch Angle Control for VariableSpeed Wind Systems," 2023 9th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore, India, 2023, pp. 559-566, doi: 10.1109/ICACCS57279.2023.10112927.

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OUTCOME IN FUTURE

The outcome and future Implications of research are vast and multifaceted

- Environmental Benefits
- Integration of Advanced Technologies
- Boost to Data-Driven Approaches
- Replicability and Scalability
- Grid Stability
- Cost Effectiveness
- Enhanced Efficiency

We will publish this research paper under the faculty of DR.M.S Minu AP/CSE; by publishing, there's a potential to set new standards for the wind energy sector, emphasizing the vital role of data analytics and adaptive control. Harmonizing these technologies can

OUTCOME IN FUTURE

revolutionize how we harness wind power, making it more efficient, reliable, and cost-effective, thereby solidifying its position as a cornerstone of our future energy landscape.